

When Compact Is Not Enough: A Protocol-Locked, Resource-Bounded Evaluation of CAPE-EEG for Expert-Vote Classification of Harmful Brain Activity

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ABSTRACT

Intensive-care EEG produces expert labels that disagree, so a useful classifier must estimate the distribution of expert opinion, stay calibrated, know when to defer, and remain auditable on modest hardware. We ask whether a 164k-parameter two-view convolutional model, CAPE-EEG, can match a small ImageNet-pretrained network when both consume the same byte-capped evidence: a *foveated* log-spectrogram of the 50-second raw EEG around each labelled window (half of the 32 time bins on the labelled centre) and a 64×64 summary of the supplied 10-minute spectrogram, stored as float16 with validity masks in a 5.58 GB cache, fused by a learned gate, with annotator disagreement as an auxiliary target. On 106,800 windows from 1,950 patients of the HMS Harmful Brain Activity dataset we froze a patient-disjoint 60/10/5/5/20 split, capped development at eight named configurations, locked the comparator, schedule and referral score in a hashed protocol document before any held-out access, and scored the test partition once. The hypothesis is falsified: three-seed patient-mean KL divergence is 0.934 for CAPE-EEG versus 0.869 for MobileNetV3-Small on identical inputs (paired difference +0.066, 95% patient-bootstrap CI [+0.033, +0.097], 391 patients). Post-lock three-seed development ablations under the frozen schedule show that foveation, the learned gate and the disagreement loss each change tune loss by less than the seed-to-seed spread, and that the comparator’s advantage is a transfer-learning effect: trained from scratch, or at half width with ImageNet weights, MobileNetV3-Small falls 0.11 and 0.10 KL *behind* the compact model. A byte-budget sweep shows that the foveation null is not an artefact of resolution: at 8, 16 and 32 local time bins the foveated and uniform encodings are indistinguishable. A patient-count learning curve confirms that the task is patient-limited, with both models still improving at the full 1,170 training patients. A Dirichlet–multinomial vote head, proposed as a principled replacement for the scalar disagreement regressor, is an equal classifier and an equal predictor of annotator disagreement, so the auxiliary signal, not its parameterisation, is the limit. The audits are the durable contribution: temperature scaling removes over-confidence in both families, entropy-based deferral lowers accepted-case risk, evidence deletion confirms that the compact model relies on the labelled raw-EEG window, the two families have different failure surfaces, and measured energy is under 3 mJ per window for both. The locked study used 0.72 GPU-hours on one desktop accelerator; code, 170 tests, executed notebooks, the protocol document and every figure with provenance are released.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health informatics*; • **General and reference** → *Reliability*.

KEYWORDS

EEG, harmful brain activity, soft labels, expert disagreement, compact neural networks, protocol-locked evaluation, calibration, robustness, resource-bounded machine learning

1 INTRODUCTION

Continuous EEG in intensive care is read for seizures and for the rhythmic and periodic patterns that surround them: lateralised and generalised periodic discharges (LPD, GPD) and lateralised and generalised rhythmic delta activity (LRDA, GRDA). Expert readers disagree about these patterns, and the largest public dataset of such annotations, HMS Harmful Brain Activity Classification, therefore scores a model against the *vote distribution* of up to 28 experts with the Kullback–Leibler divergence rather than against a single label [5, 9]. This changes what a good model is: it must be calibrated, it must know when to defer, and its use of evidence must be inspectable, because the reference itself is a disagreement.

Systems that would run such models beside a monitor face bounded memory, one shared small accelerator, and a requirement that every artefact be reproducible and inspectable. Two ideas from recent computer vision seem tailored to this regime. Foveated representations spend resolution where the label lives [1], and economical multi-scale fusion combines views without a large backbone [10]. Recent EEG work adds four warnings: preprocessing choices silently move predictions [6]; foundation models are not automatically better than task-matched controls [11]; purported evidence should be tested by deletion [17]; and robustness must be measured under declared perturbations [13].

This paper is not a leaderboard entry. Strong HMS systems such as VIPEEGNet [16] reach external KL near 0.22–0.27 on the competition’s high-agreement test set with a 12.9M-parameter pretrained backbone and ensembling; our KL is computed over *every* labelled window of a patient-disjoint split, including single-annotator windows, and is not comparable to those figures. Our question is narrower and falsifiable: under a strict, resource-bounded protocol, does a compact, evidence-selective, auditable architecture retain the discrimination of a small pretrained CNN that sees exactly the same bytes?

We answer four research questions. **RQ1 (primary)**: at a fixed cache byte budget and a bounded single-device training budget, does CAPE-EEG beat the small comparator selected on tuning data, measured by three-seed patient-mean KL on a locked, patient-disjoint test partition? **RQ2 (mechanisms)**: do temporal

foveation, a learned view gate and a disagreement auxiliary loss each improve loss at identical bytes, trunk, schedule and augmentation by more than seed variation, and does the answer for foveation depend on the byte budget? **RQ3 (confounds):** is the comparator’s advantage explained by ImageNet pretraining or by its nine-fold larger parameter count, and is the task limited by the number of training patients? **RQ4 (audits):** what do calibration, deferral, evidence-deletion, corruption and energy audits reveal that the headline KL hides, does a probabilistic vote model improve uncertainty estimation, and do the two model families fail in the same way?

Our contributions are: (1) a protocol-locked, leak-free evaluation design for expert-vote EEG classification with a patient/recording connected-component split, a 6 GB active-cache cap that counts masks and metadata, at most eight named development configurations, three fixed seeds averaged as losses, and a single scored held-out evaluation with patient-cluster bootstrap inference [2, 14]; (2) CAPE-EEG, a 164k-parameter two-view model with byte-matched temporal foveation, an inspectable learned view gate and an annotation-disagreement head, released with its exact byte- and exposure-matched controls; (3) a resolved negative answer to RQ1 and three-seed answers to RQ2 and RQ3 obtained without touching the test partition again, including a byte-budget sweep, pretraining and width controls, a patient-count learning curve and a Dirichlet–multinomial vote head; (4) audits that survive the negative result, with patient-cluster intervals throughout; and (5) an open, tested release (155 synthetic tests, seven executed notebooks, the verbatim protocol document and twenty figures with provenance sidecars) at an anonymised repository. Table 1 positions the study against the work it builds on.

2 RELATED WORK

HMS classifiers. Jing et al. [9] reached expert-level agreement on the six-pattern task, and the public HMS release [5] scores vote-distribution prediction by KL. VPEEGNet [16] adapts an ImageNet-pretrained backbone to spectrogram images and reports external KL of 0.223 and 0.273 on the competition’s agreement-weighted test protocol; those figures anchor the field but are not comparable to ours. Our comparator is the smallest member of that family, MobileNetV3-Small [7]. **Compact architectures.** Depthwise-separable convolutions [8] and Group Normalisation [19] make sub-200k-parameter trunks trainable; DPNeXt [10] motivates the exploratory multi-scale block. **Soft labels and deferral.** Learning from the full distribution of human judgments [15], temperature scaling [4] and selective classification [3] underlie the calibration and risk–coverage analyses. **Reliability of EEG decoding.** The preprocessing-sensitivity, benchmark, evidence-bottleneck and robustness studies of 2026 [6, 11, 13, 17] motivate, respectively, the STFT-resolution condition, the task-matched controls, the deletion audits and the corruption suite; none is reproduced.

3 DATA AND TASK

The source is the original HMS training release: 106,800 labelled windows from 1,950 patients, 17,089 EEG recordings (200 Hz,

19 scalp electrodes plus EKG, 50 s to 56 min long) and 11,138 spectrograms (2 s rows, four regions $\times$ 100 frequencies from 0.59 to 19.92 Hz). Each row gives an EEG offset e and a spectrogram offset s ; the labelled centre is seconds $[e+20, e+30]$ of the recording and $[s+295, s+305]$ of the spectrogram, and the two clocks differ [5]. Votes $v_{ik} \geq 0$ over six classes have totals $n_i \in [1, 28]$; 4,360 windows carry a single vote and 5,248 have tied maxima. The target is $q_{ik} = v_{ik}/n_i$. The prediction unit is a window; the independence unit is the patient, and because no recording is shared across patients the 1,950 connected components coincide with patients. Table 2 gives the deterministic allocation, which balanced rows, per-class vote mass and component counts without consulting any model.

4 CAPE-EEG

4.1 Byte-budgeted two-view evidence

Every window becomes two float16 tensors. The **local view** $L \in \mathbb{R}^{4 \times 64 \times 32}$ comes from the 50 s raw window: sixteen bipolar leads in four regional chains (LL, RL, LP, RP), per-lead median removal, a Hann STFT (256-sample window, hop 64, 153 frames), linear power averaged over the valid leads of a region, overlap-weighted integration into 64 linear frequency bins over 0.5–40 Hz and 32 time bins, then a logarithm. The *foveated* encoding covers $[0, 20]$ s with 8 bins, the labelled centre $[20, 30]$ s with 16 bins and $[30, 50]$ s with 8 bins; the *uniform* control uses 32 equal bins, at exactly the same bytes (audited with a synthetic impulse; figure in the supplement). The **context view** $C \in \mathbb{R}^{4 \times 64 \times 64}$ integrates the supplied 600 s spectrogram interval onto 64 frequency and 64 time bins per region. Each cell carries a validity bit: a frame needs at least 90% observed samples and two valid leads, a bin at least half of its weight from valid frames; non-finite supplied cells are invalid; truncated edges are masked, never shifted. One row occupies 49,152 bytes of values and 3,072 bytes of masks; the realised cache holds 106,800 rows in 5.582 GB, under the 6 GB cap, plus a 1.859 GB development-only uniform alternate. Float16 storage changes log-power by at most 0.0076. Conversion took 786 s with 16 CPU workers whose resident memory stayed below 0.77 GiB.

4.2 Architecture

Figure 1 shows the candidate. Each view has its own 3×3 stem and both share a depthwise-separable trunk with 24, 48, 64, 96 and 128 channels, frequency/time strides of (2,1), (2,2), (2,2) and (1,1), and two residual 128-channel blocks; GroupNorm avoids batch statistics. Local pooling concatenates the global mean, the global maximum and a centre-region mean whose column weights are the overlap of each bin with $[20, 30]$ s. Context pooling uses mean and maximum. Linear projections give $u_L, u_C \in \mathbb{R}^{128}$ and independent heads give p_L and p_C . The gate sees the two embeddings, the two valid fractions and the Jensen–Shannon divergence between p_L and p_C . Heads, gate and mixture run in float32 even under bf16 autocast.

4.3 Losses

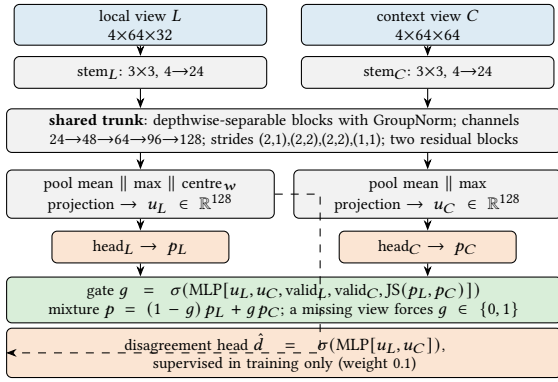
The classification loss is the soft cross-entropy $-\sum_k q_{ik} \log p_{ik}$ on the mixture, whose optimum equals that of $\text{KL}(q_i \| p_i)$. Observed pairwise annotator disagreement $d_i = 1 - \sum_k v_{ik}(v_{ik} - 1) / [n_i(n_i -$

Table 1: Positioning against related work. ✓ = present and reported; – = absent by design or not applicable; n/r = not reported in the cited description. Rows describe the cited papers as published, not their authors’ capabilities.

Study (task)	Patient-disjoint split	Byte-matched controls	Locked protocol	Soft-label KL	Separate calibration set	Deferral analysis	Evidence deletion	Declared corruption suite
VIPEGNet [16] (HMS 6-class)	n/r	–	n/r	✓	n/r	n/r	n/r	n/r
Evidence bottleneck [17] (EEG diagnosis)	n/r	–	n/r	–	n/r	n/r	✓	n/r
RobustSeiz [13] (seizure detection)	✓	–	n/r	–	n/r	n/r	n/r	✓
NeuroAtlas [11] (multi-task EEG)	✓	–	n/r	–	n/r	n/r	n/r	n/r
Preprocessing reliability [6] (EEG decoding)	✓	–	n/r	–	n/r	n/r	n/r	✓
This work (HMS 6-class)	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Frozen partitions (patient/recording components, seed 20260907). No patient, EEG or spectrogram appears in two partitions.

Partition	Rows	Patients	Share	Role
Train	64,292	1,170	0.602	development fits
Tune	10,667	196	0.100	comparator, epochs, score
Calibration-T	5,289	95	0.050	temperature only
Calibration-P	5,247	98	0.049	referral thresholds
Locked test	21,305	391	0.200	one scored evaluation

**Figure 1: CAPE-EEG (164,134 parameters).** The centre-weighted pool uses the true bin edges of the encoding in use, so the uniform control pools its real centre. Ablations remove the gate (fixed 50:50), the auxiliary head, or the foveation; no identifier, vote count or label enters the network at inference.

1)] is defined for $n_i > 1$ and not clipped to its population maximum of 5/6; the total loss is $\overline{CE} + 0.1 \cdot (\hat{d}_i - d_i)^2_{n_i > 1}$ with a differentiable zero when a mini-batch has no multi-annotator window. We note for RQ2 that d_i is a finite-sample function of the same vote vector that defines q_i (for large n_i it approaches $1 - \sum_k q_{ik}^2$), so the auxiliary target is correlated with, though not identical to, the primary target; Section 6.3 tests whether it still helps. No class weighting, vote weighting, focal loss or label sharpening is used.

4.4 Baselines and named configurations

B0 is the training-set vote prior; B1 is an L2-regularised multinomial logistic regression on 74 fixed band-power features fitted to fractional targets on the CPU. Eight named GPU configurations

were permitted in development. B2 is the byte-matched control: the same trunk on the uniform encoding with a fixed 50:50 mixture and no auxiliary head (147,524 parameters). A1 adds foveation, A2 the learned gate, P the auxiliary loss. B3 is MobileNetV3-Small [7, 18] with four input channels and six outputs, fed the two normalised views concatenated along time (64x96) and bilinearly resized to 128x192 (1,524,150 parameters). P+MSF replaces the last residual block of P by an economical multi-scale fusion block (parallel depthwise convolutions with temporal dilations 1/2/3, a 1×1 fusion and a channel gate; 208,070 parameters). The seventh and eighth named configurations, P₃ and B3₃, are P and B3 under a complete 3-epoch cosine schedule (Section 6.1). B0 and B1 need no GPU and were not counted against the cap; the permitted 0.3x learning-rate pilot was not used. Only B2 and B3 were eligible to become the comparator, chosen on tune data.

4.5 Training

All configurations use AdamW [12] (weight decay 0.01), learning rate 10^{-3} for compact models and $10^{-4}/10^{-3}$ for pretrained backbone/head with one frozen-backbone epoch, cosine decay with 200 warm-up steps, batch 64, gradient clipping at 1.0, bf16 autocast after a parity test (maximum $|\Delta p| = 0.005$), and mild augmentation only: Gaussian gain jitter, frequency masking of at most four bins, and single-region dropout with probability 0.1. Normalisation statistics (median and scaled MAD of valid cells per view and region) are fitted on the training partition only and refitted on train+tune for the final phase; invalid cells are zeroed after normalisation.

5 PROTOCOL-LOCKED EVALUATION

Figure 2 summarises the study as it ran. The design was written before the data were touched and its decisions were frozen in a hashed, timestamped protocol document (published verbatim in the repository) before any held-out access; we call this *protocol-locked* rather than pre-registered because no external registry was used. The development phase may fit at most eight named configurations on the training partition and select on tune; the only permitted optimiser pilots are a learning-rate multiplier of 0.3 and a common shorter epoch count. The lock freezes the comparator (the better of B2 and B3 on tune), the final schedule, the referral score (lowest area under the risk-coverage curve on tune among predictive entropy, one-minus-maximum-probability and $\hat{d}$), target-entropy tertiles, sixteen subgroups, six corruption conditions and the figure list. Final models are refit on train+tune for the frozen

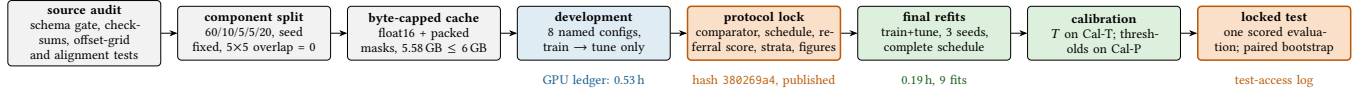


Figure 2: The protocol as executed. Every stage writes a hash (split 4455edaa, preprocess 2406ce59, cache af515c77, protocol 380269a4); a development role cannot read the test partition, and the evaluator role runs after the lock. Post-lock development analyses (Section 6.3) use only the train and tune partitions.

number of epochs with seeds 101, 202 and 303; seed 101 is the deployment exemplar chosen in advance. A scalar temperature is fitted on Calibration-T by patient-weighted cross-entropy and referral thresholds for 80/90/95% acceptance are fixed on Calibration-P. The evaluator role then reads the test partition: nine prediction passes (one per final model) and one scored evaluation. An earlier execution of the scoring script failed on a renamed dictionary key before any metric was computed; the failure and the rerun are recorded in the test-access log.

The primary estimand is $\Delta = \bar{K}_P - \bar{K}_B$, where $\bar{K}_m$ is the mean over patients of the patient-mean $\text{KL}(q||p)$ (natural log, p clipped at 10^{-7} and renormalised), averaged over the three single-model seeds—an algorithm comparison, not an ensemble. Superiority requires the 95% percentile interval of 2,000 paired patient-bootstrap replicates to lie wholly below zero. Secondary outcomes are row-weighted KL, soft cross-entropy, soft squared error, expected categorical Brier score, hard-label metrics on the 20,482 unique-majority windows (823 ties excluded), soft and majority top-label expected calibration error (ten equal-count bins, twenty-group merge rule), disagreement MSE and Spearman correlation by vote stratum, risk-coverage curves, six corruption conditions on a label-blind subset of 777 test windows from all 391 patients, view/region deletion, ranked-versus-random occlusion on 48 prespecified cases, and measured latency and memory. Sixteen subgroup contrasts are exploratory and reported without multiplicity adjustment. All GPU wall time is charged to a 12-hour ledger.

6 RESULTS

6.1 Development phase and the schedule finding

The development phase (supplementary Table S1) proceeded as follows. Deep models beat the CPU feature baseline by 0.25 KL. Every network reaches its best tune loss after one to three epochs and then memorises patient-specific structure: with a median of 19 windows per patient, 64k rows behave like far fewer independent examples. Because a final refit must run a complete schedule without checkpoint selection, the two remaining named configurations were spent on a complete 3-epoch cosine schedule; it improved P from 0.983 to 0.9270 and gave B3 0.9266. These are two distinct runs with distinct prediction files, not a copy. The comparator rule was fixed in advance: the better of B2 and B3 on tune. B3 was the only eligible control at the frozen schedule and was also better than B2 by 0.090 at the original schedule, so it became the comparator; P is the candidate by definition and never competes for the comparator role. Entropy was frozen as the referral score (tune AURC 0.402 versus 0.407 for one-minus-maximum and 0.436 for $\hat{d}$).

6.2 RQ1: primary confirmatory result

The hypothesis of RQ1 is rejected (Figure 3a, Table 3). With calibrated predictions $\Delta = +0.0657$, 95% CI $[+0.0334, +0.0973]$: the candidate is inferior. The uncalibrated secondary estimate agrees $(+0.0485, [+0.0059, +0.0899])$. Per-seed calibrated patient-mean KL is 0.924, 0.955 and 0.924 for P and 0.878, 0.856 and 0.872 for B3; 35.3% of patients are better predicted by the candidate. The paired standard deviation on the 391 test patients is 0.329, smaller than the development estimate of 0.58 because the latter was computed on 196 tune patients with checkpoint-selected 12-epoch models; at the observed spread the minimum detectable difference at 80% power is 0.047, below the observed effect, so the negative result is resolved rather than under-powered. B3 leads on every secondary metric, including macro AUROC (0.884 versus 0.857) and macro AUPRC (0.656 versus 0.580). LRDA is the hardest class for both models (per-class AUROC 0.80–0.85, recall below 0.32).

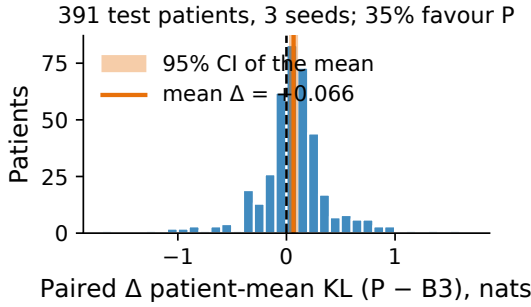
6.3 RQ2 and RQ3: mechanism ablations and confound controls

The single-seed development ladder (supplementary Table S1) suggested that foveation (+0.015), the learned gate (−0.011) and the disagreement loss (−0.007) barely move tune loss. Because those deltas are smaller than the seed spread observed later in the final phase, we re-ran the ladder after the lock on development data only, with three seeds under the frozen schedule (Table 4, Figure 3b). The answer to RQ2 is that none of the three mechanisms produces a benefit that exceeds seed variation: foveation changes tune KL by −0.001 (± 0.035), the gate by +0.011 (± 0.036) and the auxiliary loss by −0.000 (± 0.056); together they give +0.010 (± 0.056). We therefore make no mechanism claim in either direction. The exploratory multi-scale block is significantly worse than P on the locked test (+0.024, $[+0.001, +0.045]$).

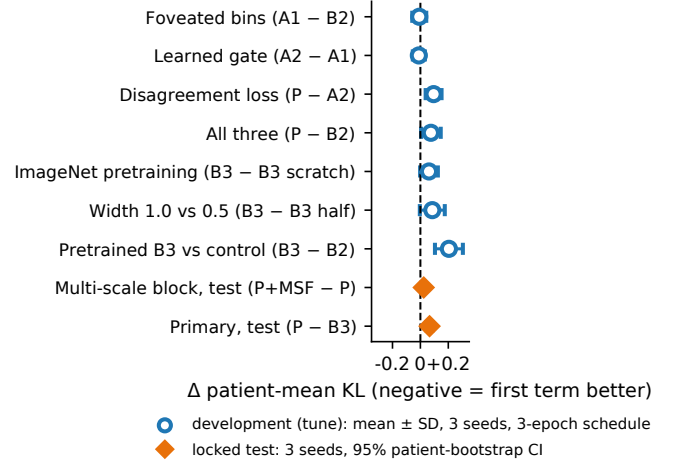
For RQ3 we trained the comparator from random initialisation (B3-scratch) and at half width with ImageNet weights (B3-half) under the same schedule and seeds. ImageNet pretraining is worth −0.250 (± 0.086) tune KL and full width −0.193 (± 0.040). Without either, the MobileNet is no longer ahead: B3-scratch is +0.160 (± 0.060) worse than P and the 575k-parameter B3-half is +0.102 (± 0.051) worse, on every seed for the former. The comparator’s advantage is therefore the combination of transferred features and the width those features were learned at, not a property of the inverted-residual trunk itself; a compact trunk trained from scratch on 1,170 patients is competitive with, and more stable than, an equally scratch-trained MobileNet. The candidate’s own tune spread under this schedule is 0.046, the largest of the seven, and Section 6.7 traces part of it to the gate.

Table 3: Locked-test metrics, three-seed means. “Cal” = scalar temperature from Calibration-T (rank-based metrics are unchanged by a positive temperature). Hard-label metrics use the 20,482 unique-majority windows. Lower is better for KL, CE, Brier and ECE.

Method	Params	Patient KL	Row KL	KL _{≥10} votes	Soft CE	Brier	Macro AUROC	Macro AUPRC	Bal. acc.	Macro F1	MCC	Soft ECE
B3 raw	1,524,150	0.943	0.869	0.754	1.230	0.592	0.884	0.656	0.576	0.580	0.509	0.107
B3 cal		0.869	0.807	0.666	1.168	0.578	0.884	0.656	0.576	0.580	0.509	0.036
P raw	164,134	0.991	0.961	0.819	1.322	0.642	0.856	0.579	0.523	0.520	0.446	0.118
P cal		0.934	0.906	0.744	1.267	0.625	0.857	0.580	0.523	0.520	0.446	0.051
P+MSF cal	208,070	0.958	0.925	0.738	1.286	0.634	0.849	0.561	0.506	0.502	0.428	0.040



(a) Primary result on the locked test: per-patient paired differences in calibrated patient-mean KL (P – B3, three-seed mean) and the 95% bootstrap interval of their mean.



(b) Ablation ladder. Circles: post-lock development contrasts (tune, mean $\pm$ SD over three seeds, frozen schedule). Diamonds: locked-test contrasts with 95% patient-bootstrap intervals.

Figure 3: Confirmatory result and mechanism ladder.

Table 4: Post-lock development ablations under the frozen complete 3-epoch schedule (last-epoch weights, no checkpoint selection): tune patient-mean KL, mean $\pm$ SD over seeds 101/202/303. Paired within-seed contrasts are in Figure 3b and the supplement. Test data were not used.

ID	Configuration	Params	Tune KL (mean $\pm$ SD)
B2	uniform, fixed fusion, no aux	147,524	0.980 $\pm$ 0.014
A1	+ foveated bins	147,524	0.979 $\pm$ 0.025
A2	+ learned gate	155,877	0.991 $\pm$ 0.017
P	+ disagreement loss	164,134	0.990 $\pm$ 0.046
B3	MobileNetV3-Small, ImageNet	1,524,150	0.899 $\pm$ 0.025
B3-scratch	same, random init	1,524,150	1.150 $\pm$ 0.076
B3-half	width 0.5, ImageNet	574,518	1.092 $\pm$ 0.016

Inference-time modality ablation of the frozen P model on the full test (RQ4) confirms that the design uses what it claims: forcing the context path only (local view removed) raises calibrated patient-mean KL by +0.408 [+0.35, +0.47] and flips 31.5% of labels, whereas forcing the local path only costs +0.090 [+0.06, +0.13]; region removal costs +0.10 (LL), +0.12 (RL), +0.20 (LP) and +0.07 (RP) (Figure 6a). The gate’s test mean is 0.45, 0.72 and 0.43 for the three seeds, so the mixture is real but seed-dependent, which is

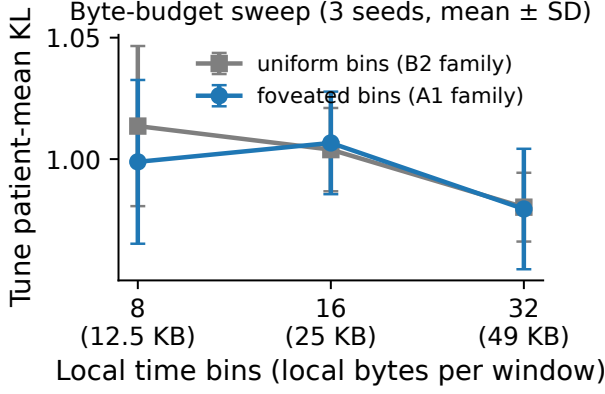
one source of the candidate’s larger seed spread (SD 0.018 versus 0.011).

6.4 RQ2 continued: byte-budget sweep

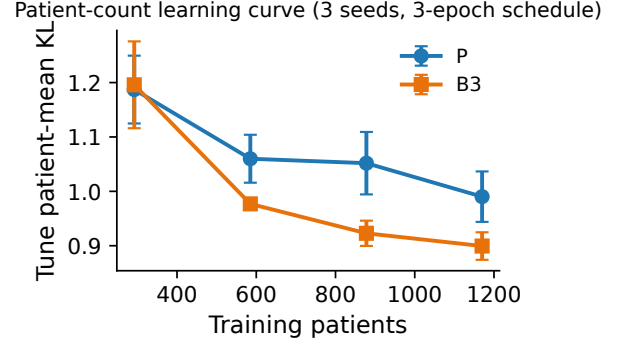
A null at 32 bins could be specific to that budget: uniform bins of 1.56 s already resolve the 10-second centre, so foveation has little to redistribute. We therefore re-ran the fixed-fusion pair (B2 versus A1 families) at 16 and 8 local time bins, halving and quartering the local bytes, with three seeds under the frozen schedule (Figure 4a). The paired foveated-minus-uniform difference is -0.001 ($\pm$ 0.035) at 32 bins, +0.003 ($\pm$ 0.004) at 16 and -0.015 ($\pm$ 0.054) at 8. Tune KL moves only from 0.980 (uniform, 32) to 1.014 (uniform, 8): most of the discriminative evidence at this resolution is spectral rather than fine-temporal, and foveation is not rescued by a tighter budget in this range.

6.5 RQ3 continued: patient-count learning curve

To test the claim that the task is patient-limited, we trained both final architectures on deterministic 25, 50 and 75% subsets of the training patients (Figure 4b). Tune KL falls from 1.187 to 0.990 for P and from 1.196 to 0.899 for B3 as patients grow from a quarter to



(a) Byte-budget sweep: coarser encodings are exact unions of the cached 32-bin cells.



(b) Patient-count learning curve on 25/50/75/100% of training patients.

Figure 4: Post-lock development analyses (tune partition, three seeds, frozen 3-epoch schedule, mean $\pm$ SD).

the full 1,170, with fitted slopes of -0.092 and -0.151 KL per doubling of patients; neither curve has flattened. The comparator's margin is absent at a quarter of the patients (1.187 versus 1.196) and opens as patients are added, so transferred features are exploited better by more patients rather than being a small-data advantage, and the compact model's larger seed spread persists throughout. More patients, not more windows per patient, is the resource that would move both models.

6.6 RQ4: a Dirichlet-multinomial vote head (CAPE-EEG v2)

The scalar disagreement target d_i is a finite-sample function of the same votes that define q_i , which limits what the auxiliary head can add. A principled alternative models the vote-generation process itself: $\mathbf{v}_i \sim \text{DirMult}(n_i, c(x_i) \mathbf{p}(x_i))$, where $\mathbf{p}$ is the mixture probability already produced by the network and $c(x) > 0$ is a learned concentration, low when annotators are expected to disagree. We implemented this as P-DM: the auxiliary head outputs $\log c$ through a softplus, the loss adds $0.1 \times$ the per-vote negative log-likelihood of the observed counts (normalised by n_i so that heavily annotated windows carry no extra weight), and the implied pairwise disagreement $\hat{d} = 1 - \sum_k p_k (cp_k + 1) / (c + 1)$ is reported like the scalar head's output. On development data with three seeds under the frozen schedule, P-DM reaches tune KL 0.997 (± 0.044) against 0.990 (± 0.046) for P and 0.991 (± 0.017) for A2 without any auxiliary head (paired P-DM - P: $+0.007 \pm 0.043$). Its implied disagreement correlates with observed annotator disagreement at Spearman 0.310, the same as the scalar head's 0.313; as a referral score its AURC is 0.472 against 0.483 for the scalar head and 0.450 for predictive entropy on the same predictions. The probabilistic formulation is better justified and gives a concentration that can be read directly as expected annotator agreement, but it does not extract more from the votes than the scalar head: the limit is the information in the auxiliary signal, which is largely a function of the soft label, not its parameterisation. Neither head changes the answer to RQ1.

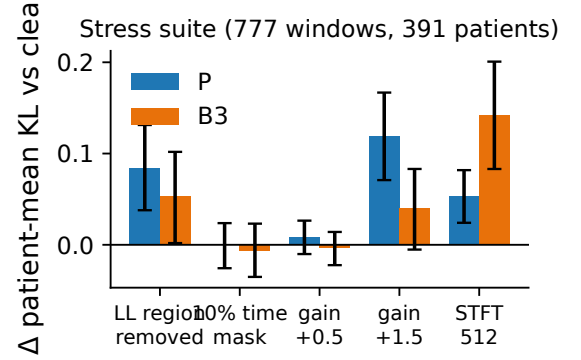


Figure 5: Six-condition stress suite (777 label-blind test windows, 391 patients): change in calibrated patient-mean KL versus the clean input with paired patient-bootstrap intervals.

6.7 RQ4: calibration, deferral and disagreement

Both architectures are over-confident when trained with soft cross-entropy: temperatures fitted on Calibration-T range from 1.21 to 1.39 (P) and 1.28 to 1.33 (B3), none at a bound, and scaling cuts soft top-label ECE from 0.10–0.14 to 0.03–0.06 and patient-mean KL by 0.046–0.081 for every model and seed (reliability and risk-coverage curves in the supplement). With entropy as the frozen score and thresholds fixed on Calibration-P, referring 11.8% of test windows lowers P's accepted-case patient KL from 0.923 to 0.911 (seed 101); for B3, referring 9.1% lowers it from 0.878 to 0.874. The benefit is modest; AURC is 0.401–0.415 for P against 0.365–0.372 for B3. The disagreement head predicts observed annotator disagreement with Spearman 0.25–0.31 across seeds, but as a referral score it is worse than entropy (AURC 0.444–0.448).

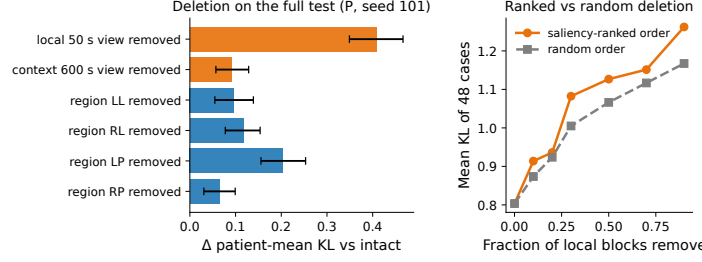


Figure 6: Evidence-use audit of the frozen P model (seed 101). (a) View and region deletion on the full test with patient-bootstrap intervals. (b) Saliency-ranked versus random deletion of local (region, time-bin) blocks on 48 prespecified cases.

6.8 Subgroups, robustness and evidence use

Across the sixteen frozen strata (exploratory; author version appendix and repository), the comparator’s advantage concentrates in confident windows and in LPD: $+0.079$ [$+0.027$, $+0.129$] for windows with at least ten votes, $+0.073$ [$+0.028$, $+0.116$] for unanimous windows, $+0.179$ [$+0.046$, $+0.308$] for LPD-majority windows and $+0.119$ [$+0.029$, $+0.206$] for windows with fewer than 95% valid cells; on high-entropy windows ($+0.019$, $[-0.033$, $+0.071]$), LRDA (-0.028 , $[-0.154$, $+0.093]$), GPD, GRDA, seizure and tied windows the two models are indistinguishable, and single-vote windows have low support (28 patients).

Figure 5 reports the corruption suite. Both models are insensitive to the additional time mask and to the mild gain shift. The candidate degrades three times more under a strong amplitude shift ($+0.119$ versus $+0.040$), whereas the pretrained network degrades nearly three times more under a change of STFT resolution from 256 to 512 samples ($+0.142$ versus $+0.053$), with 30% of its labels flipping: the two families have different failure surfaces, and the second pattern is the one the preprocessing-reliability literature predicts [6]. Figure 6b shows that deleting local blocks in order of $|\nabla_x \odot x|$ raises KL faster than a random order at every fraction (1.08 versus 1.01 at 30%), so the model’s importance ranking is informative; neither fact makes it accurate.

6.9 Resources and latency

The locked study consumed 0.722 GPU-hours in 17 ledgered jobs with no failures (baselines 0.171 h, ablations and pilots 0.362 h, nine final fits 0.188 h); the 19 post-lock development runs added 0.36 h, for 1.08 GPU-hours in total. A final fit takes 1.11 min (P) or 0.98 min (B3) on the GB10 with a peak CUDA allocation of 0.44 GiB and a process resident set of 7.6 GiB. Cached-tensor inference has median latency 1.36 ms at batch 1 and 2.62 ms at batch 32 for P, and 1.50 ms and 1.87 ms for B3: the batch-1 margin is within measurement noise, and at batch 32 the fused MobileNet kernels are faster despite nine times more parameters. Raw Parquet-to-prediction latency is about 45 ms per window for both models and is dominated by I/O and the STFT. Sustained batch-32 inference draws 41.4 W for P and 46.9 W for B3 above an idle package power of 12.3 W (nvidia-smi, 10 Hz sampling over 20 s), which is 2.941 and 2.269 mJ of incremental energy per window (4.19 and 3.08 mJ including idle);

the comparator is the more energy-efficient of the two on this accelerator because of its higher throughput. Without any accelerator, single-thread CPU inference on the host’s ARM cores takes 8.8 ms per window for P and 10.8 ms for B3 (float32, batch 1; four threads are slower for both because of threading overhead at this size). These are desktop-class measurements with a stated boundary, not embedded-device measurements.

7 IMPLEMENTATION AND REPRODUCIBILITY

The cache converter writes float16 values and packed masks into pre-allocated memory-mapped shards through atomically renamed .partial files; a supervisor samples CUDA, process-tree and system memory every fifty steps and aborts at declared stops; a flock lock forbids concurrent GPU jobs; a development-role loader raises on any request for the test partition; and the release is an allowlisted export with identifier and secret scans of the working tree and every Git object. The repository ships 170 synthetic CPU tests, eight executed notebooks, the protocol document and every figure with a provenance sidecar; engineering details are in the supplement.

8 DISCUSSION

Answers. RQ1: no; the candidate is inferior by 0.066 patient-mean KL with an interval wholly above zero. RQ2: none of the three mechanisms improves loss beyond seed variation under the frozen schedule, and foveation stays null at a quarter of the local byte budget. RQ3: the comparator’s advantage is a transfer-learning effect at full width; without ImageNet weights, or at half width, MobileNetV3-Small falls behind the candidate; and both models are still learning at the full patient count, so the task is patient-limited. RQ4: the audits show two model families that are both over-confident before scaling, both improved by entropy-based deferral, fragile to different perturbations, and cheap to run (under 3 mJ per window); the compact model demonstrably relies on the labelled raw-EEG window, and a Dirichlet–multinomial head neither improves loss nor disagreement prediction over the scalar head, so the auxiliary signal itself is the limit.

Why the compact candidate lost. The task is patient-limited: 1,170 training patients contribute 64k highly correlated windows and every architecture overfits within three epochs. Under the frozen schedule the candidate ties the comparator on tune but not

on the larger test, and its gate makes its seed spread 60% larger. ImageNet features transfer to the resized two-view log-spectrogram image and are worth 0.19 KL on tune; stripped of them, or at half width, the MobileNet falls behind the compact trunk trained from scratch. The comparator's advantage is largest where evidence is clearest (unanimous or heavily annotated windows, LPD), which is where transferred texture detectors help most. The three mechanisms address problems the data do not pose: at 32 total bins the uniform encoding already resolves the 10-second centre at 1.56 s, so the foveation null may be specific to this byte budget; the two views mostly agree, so the gate re-weights similar distributions; and the disagreement target is largely a function of the soft label already being fitted.

Compactness is not efficiency. A $9\times$ smaller parameter count bought no throughput, energy or batch-1 latency advantage on the accelerator, while costing accuracy and calibration against the pre-trained network; it did buy accuracy and stability against the same network trained from scratch. For resource-bounded clinical inference, measured latency, memory and robustness under the deployment's own preprocessing are the quantities to report; parameter count is a poor proxy.

Threats to validity. Only two configurations were eligible as comparator, so a stronger small network could widen the gap; the post-lock ablations are development-only and cannot be confirmatory; calibration and referral thresholds were fitted on 95 and 98 patients; the lock is self-attested through hashes and logs rather than an external registry; subgroup contrasts are unadjusted for multiplicity.

Limitations. Single dataset and site; segment labels only (no event-level sensitivity, onset latency or false-alarm rate); exchangeable-annotator assumption behind d_i ; representation-level corruptions; noisy single-annotator targets; weights withheld pending a rights review; desktop-class latency and energy. The one improvement this study could not make is external validation on a second corpus: the only public EEG corpus with periodic-discharge labels (TUEV) requires registered access, so the repository ships a fixed label mapping, alignment rule and lock procedure for it (docs/09) rather than a result. Natural next hypotheses under a new lock are patient-level regularisation and self-supervised pre-training of the compact trunk on the unlabelled recordings.

9 CONCLUSION

We asked whether byte-matched temporal foveation, a learned view gate and an annotation-disagreement auxiliary target let a 164k-parameter model match a small pretrained CNN on expert-vote EEG classification, and we specified how the question would be answered before touching the data. The answer is no. Post-lock three-seed controls show that none of the mechanisms helps beyond seed variation, and that the comparator wins through ImageNet transfer at full width rather than through its architecture: trained from scratch or at half width it falls behind the compact model. The protocol, the audits and the resource accounting are the reusable result: a complete, leak-free, audited evaluation of a clinical EEG model fits in about one GPU-hour on a desktop accelerator, and the next attempt can start from the published

lock. Code, tests, executed notebooks, the protocol document and every aggregate table are available at an anonymised repository.

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